

Hedging Bitcoin with Its Own Fear Gauge

Evidence from the BVIV Perpetual on Volatility Forecasts, Funding Costs and 15-Minute VWAP Timing

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Abstract

Bitcoin implied volatility now has a perpetual: a perpetual future on the Volmex BVIV index has traded on Bitfinex since April 2024. We use 15-minute data from 1 January 2023 to 3 October 2026, including the perpetual's complete Bitfinex record, to ask how BTC volatility should be forecast when a 30-day implied index is observable, and whether a long-BVIV position, possibly timed by a volume-weighted average price (VWAP) signal on 15-minute bars, hedges BTC. HAR augmented with implied variance is the best one-day forecast, significantly better than HAR; at a week and a month HAR-IV and the bias-corrected index lead, but not significantly. With rules frozen on simulated training data and executed one bar after the signal at the mark, a minimum-variance BVIV hedge reduced the expected shortfall (ES) of a long-BTC book by 17.9% (95% block-bootstrap CI 6.2–32.4%), mostly on a handful of crash days in 2026. Funding settled at its $\pm 0.25\%$ cap in 41% of 8-hour periods and cost a permanently long contract 24 vol points a year; the perpetual traded a median \$71 a day, and on its best days the hedge of a single bitcoin exceeded the market's open interest. A VWAP ratchet added +2.37 points of ES reduction (CI $[-0.22, +4.89]$), but placebos with the same triggers at shifted dates captured most of it: the gain is mainly hedge size, with timing worth +0.33 points. A calibrated simulation with an arbitrage-free perpetual reproduces the forecast ranking and the split between size and timing. BVIV perpetuals worked as a crash hedge, but funding, execution speed and capacity, not signal design, decide what that hedge is worth.

1 Introduction

A holder of bitcoin, in spot or through a perpetual swap, faces two kinds of risk: an ordinary random walk, and sudden crashes in which liquidity disappears and implied volatility spikes. Futures hedges remove the first and, with it, the position itself; crash protection has required options. The Volmex BVIV index measures 30-day BTC implied volatility in a model-free way (Volmex Labs, n.d.), and it has traded as a USDT-margined perpetual on Bitfinex since April 2024 (CoinDesk, 2024), on gTrade since June 2025 (CoinDesk, 2025a) and on Hyperliquid since September 2026 (CoinDesk, 2026). CME announced Bitcoin Volatility futures in May 2026 (CME Group, 2026).

Whether long volatility hedges long BTC depends on the spot-vol correlation, which in BTC is weak and unstable. Positive return shocks have historically raised volatility more than negative ones (Bouri et al., 2017; Baur and Dimpfl, 2018), an asymmetry that has faded (Takaishi, 2021); the BTC–DVOL correlation flipped positive in early 2023 (CoinDesk, 2023), while BVIV trended down through the 2024–25 rally (CoinDesk, 2025b). Equity investors learned that long-volatility hedges look good *ex ante* but are eroded by carry and costs (Whaley, 2013; Alexander et al., 2016; Israelov, 2019), except in genuine crises (Szado, 2009). This raises two questions: what does such a hedge cost on a real crypto venue, and can it be *timed* by a signal every intraday trader watches, price relative to its VWAP?

Contributions. (i) We provide, to our knowledge, the first evidence on hedging BTC with a tradable implied-volatility perpetual, including an exact reconstruction of its funding rule, its realised costs and its capacity. (ii) We compare seven volatility forecasts on the official BVIV index. (iii) We test VWAP timing with rules frozen on simulated data, single-path block-bootstrap inference and a *timing placebo* that splits a rule's edge into hedge size and timing. (iv) A calibrated simulation with an arbitrage-free perpetual serves as a controlled experiment. A pipeline regenerates every number from public data, except one dated order-book snapshot.

2 The BVIV index and its perpetual

Index. BVIV is a 30-day constant-maturity implied-volatility index that Volmex describes as model-free (Volmex Labs, n.d.); secondary sources list Deribit and OKX options as inputs, and Volmex serves it at

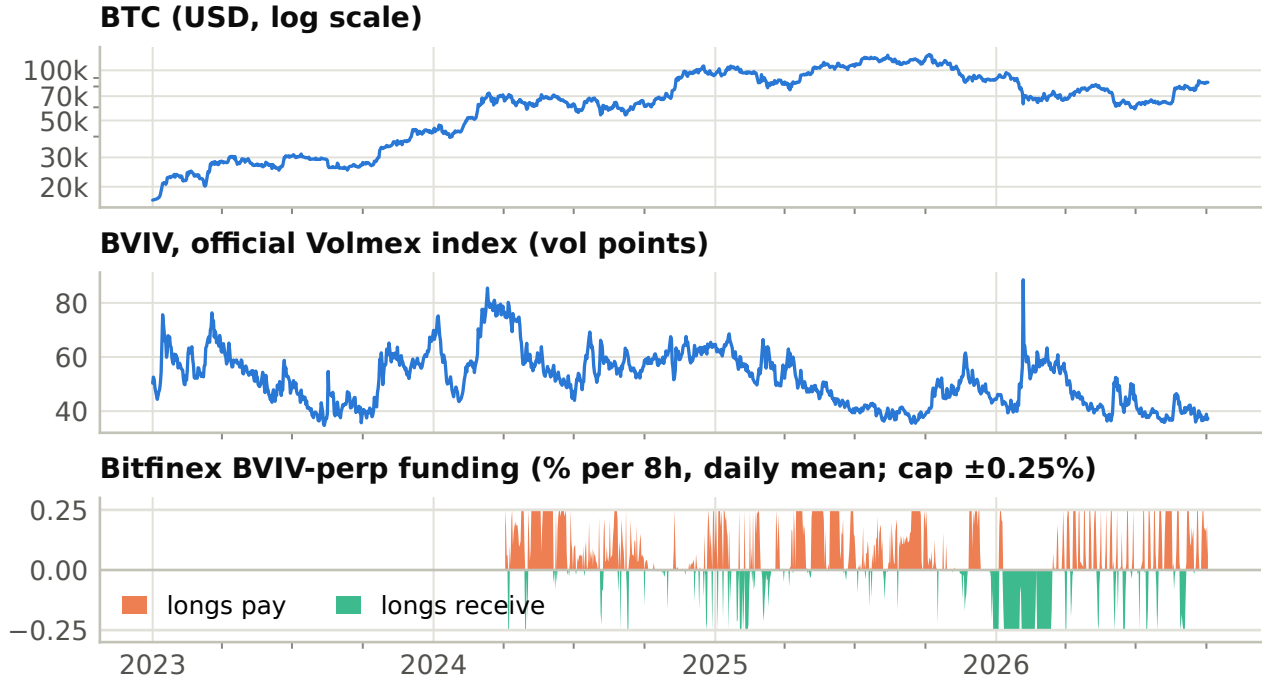


Figure 1: Live data, 1 January 2023 to 3 October 2026. Top: BTC (Binance). Middle: the official Volmex bviv index. Bottom: Bitfinex bviv-perpetual funding per 8-hour period (daily mean), paid by longs when positive.

15-minute resolution. Deribit’s DVOL applies a VIX-style variance-swap calculation (Demeterfi et al., 1999) to Deribit options alone (Deribit, 2021). Implied variance exceeds subsequent realised variance on average: Almeida et al. (2024) estimate about 0.14 a year in variance units for 2017–22, consistent with Alexander and Imeraj (2021) and the equity evidence of Carr and Wu (2009).

The Bitfinex perpetual. One contract pays one dollar (USDt) per index point; the mark price is the index converted at USD/USDt and deviates from the official Volmex index by at most 0.15% on 95% of 15-minute bars. Funding settles every eight hours. The status feed exposes the accrued premium a of the perpetual over its mark, and the settled rates follow $\text{sign}(a) \min\{\max(|a| - 0.05\%, 0), 0.25\%\}$ exactly (largest deviation 0.05 bp over all 2,740 settlements). The cap, 0.75% of notional a day, binds often: in 41% of settlements from April 2024 to October 2026, and longs paid on 52%. A permanently long contract paid 24 vol points a year on average (51% of notional), but the burden swung from 46 points in 2024 and 34 in 2025 to -12 in 2026, when longs *received* funding (Figure 1). The market is tiny. The perpetual traded on 576 of 914 days, with a median daily volume of \$71 (mean \$7,300), and only 4.4% of 15-minute bars saw a trade. Its quoted mid sat a median 0.05% above the mark, but dust orders at the top of the book put it more than 5% away on 1.4% of bars (up to +276%), so it is not an executable price.

3 Data and volatility forecasts

Data. BTC is the Binance BTCUSDt spot market on 15-minute bars (1372 days); quote volume over base volume gives each bar’s exact VWAP. bviv is the official Volmex series from its public 15-minute history. The perpetual’s marks, quotes, open interest and accrued funding come from about 1.3 million Bitfinex status snapshots (one a minute), which also yield every funding settlement; trades come from Bitfinex’s trade candles, and BTC-perpetual funding for the perpetual-book variant is Bitfinex’s own. Table 1 summarises the sample. Volatility averaged 46.6% against a mean index of 51.9 (daily closes 34.7–88.5); implied exceeded subsequent realised volatility on 75% of days. The daily spot-vol correlation was -0.10 overall but moved from $+0.32$ in 2023 through $+0.03$ and -0.30 to -0.59 in 2026: the market became VIX-like. The index lags BTC: its 15-minute changes are autocorrelated (0.20), and 41% of its response to a 15-minute BTC return arrives in the three following bars, which matters for execution (Section 4).

Forecasts. From 15-minute log returns we compute daily realised variance (Andersen et al., 2003); fifteen minutes is coarser than the 5-minute rule of thumb (Liu et al., 2015) but matches the hedge’s frequency. Each model forecasts the average daily variance over the next $h \in \{1, 7, 30\}$ days from information at day t ’s close: a random walk, RiskMetrics EWMA (J.P. Morgan/Reuters, 1996), GARCH(1,1) with Student- t errors

Moment	Live	2023	2024	2025	2026	Sim.
Realised vol, annualised (%)	46.6	43.2	53.0	44.6	44.3	51.5
Mean BVIV (vol pts)	51.9	52.4	60.2	48.3	45.2	57.1
IV – subsequent 30d RV (vol pts)	6.8					7.6
Share of days IV > RV (%)	75					78
corr(daily return, ΔIV)	−0.10	0.32	0.03	−0.30	−0.59	−0.21
corr on falling 15m bars	−0.24					−0.26
corr on rising 15m bars	0.03					0.03
AC(1) of 15m index changes	0.20					−
Excess kurtosis, daily returns	3.4					3.7
Worst daily log return (−%)	15.1					15.0

Table 1: Stylised facts: live sample (1 January 2023 to 3 October 2026) and by calendar year, against the median of the simulation’s 200 test paths (Section 6).

Model	$h = 1$ day			$h = 7$ days			$h = 30$ days		
	QL ratio	MZ R^2	DM t	QL ratio	MZ R^2	DM t	QL ratio	MZ R^2	DM t
RW	1.71	0.16	10.47	1.33	0.15	2.56	1.42	0.09	1.70
EWMA	1.27	0.09	5.18	1.40	0.10	2.91	1.87	0.09	2.53
GARCH	1.41	0.10	8.96	1.95	0.09	5.29	3.52	0.14	3.21
HAR	1.00	0.21	−	1.00	0.19	−	1.00	0.02	−
IV-raw	1.24	0.15	6.07	1.06	0.22	0.79	0.91	0.26	−0.65
IV	1.17	0.15	2.72	1.02	0.20	0.15	0.84	0.18	−1.54
HAR-IV	0.96	0.26	−2.25	0.95	0.25	−0.74	0.92	0.16	−0.60

Table 2: Live out-of-sample forecast accuracy, 2024 to October 2026. QL ratio: a model’s QLIKE over HAR’s (best in bold). DM t : Diebold–Mariano statistic against HAR (negative favours the model).

re-estimated monthly, the heterogeneous autoregressive (HAR) model of Corsi (2009) with a 7-day week, the index itself, raw ($IV_t^2/365$, “IV-raw”) and bias-corrected (“IV”), and HAR augmented with implied variance:

$$\ln \bar{RV}_{t+1:t+h} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln \bar{RV}_{t-6:t} + \beta_m \ln \bar{RV}_{t-29:t} + \beta_{iv} \ln(IV_t^2/365) + \varepsilon_{t+h}. \quad (1)$$

HAR and IV are the special cases without the implied or the realised terms. Regressions run in logs, so forecasts $\exp(\hat{y} + s^2/2)$ are positive, and are re-estimated daily on an expanding window of targets already observed. We evaluate the 1,006 one-day forecasts from 2024 on with QLIKE, which is robust to proxy noise (Patton, 2011), the Mincer–Zarnowitz R^2 and Diebold–Mariano (DM) tests against HAR with Newey–West variance (Mincer and Zarnowitz, 1969; Diebold and Mariano, 1995).

Results. Table 2 confirms the pattern documented on Deribit data by Hoang and Baur (2020). At one day HAR-IV is best (QLIKE ratio 0.96) and significantly better than HAR (DM -2.25), while the index alone is worse (1.17) and GARCH on daily returns lags (1.41), as Bergsli et al. (2022) found for BTC. At longer horizons the leading models (HAR, HAR-IV and the index) are not significantly different. HAR-IV stays ahead at seven days (0.95; DM -0.74), where the bias-corrected index is no better than HAR (1.02); at 30 days the bias-corrected index is best (0.84; DM -1.54), followed by raw IV (0.91) and HAR-IV (0.92). The calibrated simulation has the same winner at every horizon (Section 6). The 30-day HAR-IV forecast feeds the hedge sizing below.

4 Hedging framework

P&L. The book is long q BTC at price S_t and long $H_t \geq 0$ bviv-perpetual contracts with mark F_t . A position chosen at the close of 15-minute bar t earns over bar $t+1$

$$\begin{aligned} \Pi_{t+1} = & q(S_{t+1} - S_t) - \mathbb{1}_{\text{perp}} q S_t f_t \Delta t + H_t(F_{t+1} - F_t) - H_t \psi_t \\ & - (H_{t+1} - H_t)(P_{t+1} - F_{t+1}) - c |H_{t+1} - H_t| F_{t+1}, \end{aligned} \quad (2)$$

where $\mathbb{1}_{\text{perp}} = 1$ for a perpetual BTC book, Δt is the bar length in years, f_t the BTC perpetual’s annualised funding rate, ψ_t the bviv funding per contract, booked on the position held at each settlement, P_t the fill price, and c a proportional cost of 6 bp fee plus 10 bp slippage per side. On live data ψ_t is the settled funding. A perpetual on an index I_t is priced at the expected index sampled at a random time (He et al., 2022; Ackerer et al., 2025); in the simulation we split its funding into an arbitrage-free drift transfer $\mathbb{E}_t[I_{t+1}] - I_t$ and a carry premium.

Execution. On live data a decision at the close of bar t is executed at the close of bar $t+1$ at the mark ($P = F$), plus costs. Same-bar execution would collect the index’s predictable catch-up after a BTC move (Section 3), and the quoted mid is not executable (Section 2); both are sensitivities (quotes capped at $\pm 5\%$ of the mark), as are costs of 50 bp per side, what the book charged for one bitcoin’s hedge at the end of the sample.

Hedge ratios and signal. The minimum-variance hedge is $H_t^* = -qS_t\beta_t$, with $\beta_t = \text{Cov}_t(r, \Delta F) / \text{Var}_t(\Delta F)$ estimated on 15-minute changes by an EWMA (half-life 14 days) (Johnson, 1960; Ederington, 1979); we impose $H \geq 0$. A downside ratio $\beta_t^- = \mathbb{E}_t[r \Delta F \mid r < 0] / \mathbb{E}_t[\Delta F^2 \mid r < 0]$ on falling bars, shrunk towards β_t with a two-day prior, gives the crash hedge H_t^- . The breakdown statistic compares the price with its rolling 24-hour VWAP,

$$z_t = \ln(S_t / \text{VWAP}_t^{24h}) / (\hat{\sigma}_t \sqrt{1/3}), \quad (3)$$

because for Brownian prices the gap between the last price and its average over a window w has standard deviation $\sigma\sqrt{w/3}$. Here $\hat{\sigma}_t$ is the daily volatility implied by a causal EWMA (half-life one day) of 15-minute squared returns, de-seasonalised by a trailing hour-of-week profile (Andersen and Bollerslev, 1997; Hansen et al., 2024).

Rules. All rules are long-only, their targets are capped at 50% of BTC notional, and they trade only when the target leaves a 25% band around the position (Whalley and Wilmott, 1997), so held positions can drift above the cap (at most 64% on live data). *Always-on* holds λH_t^* with $\lambda \in \{1, 1.5\}$. The *VWAP-switch* holds H_t^- while a gate is on, switching on at $z_t < z_{in}$ and off at $z_t > z_{out}$, for at least four hours. The *VWAP-ratchet* holds a core ϕH_t^* plus an overlay o_t that jumps to 1 on a breakdown and decays with half-life τ :

$$o_t = \begin{cases} 1 & z_t < z_{in} \\ 2^{-1/(96\tau)} o_{t-1} & \text{otherwise,} \end{cases} \quad H_t = \phi H_t^* + o_t \max(H_t^- m_t - \phi H_t^*, 0), \quad (4)$$

where $m_t = \min\{\max(e^{c_t}, 0.5), 1.5\}$ optionally scales the overlay by forecast “cheapness” c_t : the log ratio of the 30-day HAR-IV forecast to implied variance, less its trailing 180-day median, lagged one day.

Protocol. Rule parameters were selected on 48 simulated training paths (Section 6), out of 72 ratchets and six switches, by median ES reduction net of costs, before the live pipeline was written. The selected ratchet keeps the full core ($\phi = 1$), triggers at $z < -1.5$, decays with a half-life of 4 days and uses forecast sizing; the selected switch enters at $z < -1$ and exits at $z > 0.5$. The live sample is therefore out of sample for the rules, which avoids data snooping (White, 2000; Hansen, 2005; Bailey et al., 2014), although the simulator’s calibration targets include coarse features of the 2023–26 market. Signals warm up on 2023 data, and hedging is evaluated over the 914 days since the listing, 3 April 2024 to 3 October 2026. On a single path we report 95% intervals from a paired stationary block bootstrap over days (mean block ten days, 2,000 draws); for mean blocks from one to 40 days the lower bound of the always-on ES reduction lies between 4.7% and 7.3%. The timing placebo shifts a ratchet’s trigger series circularly within the evaluation window, by every multiple of a week for the selected rule (129 shifts) and of two weeks for each rule of the grid (64 shifts). It keeps the triggers’ number, clustering and hour-of-week profile but breaks their link to the market. A ratchet’s edge over the always-on hedge then splits into *size*, the placebo mean minus the always-on result, and *timing*, the ratchet minus its placebo mean.

5 Results on live data

The hedge works, in crashes. The minimum-variance hedge reduced ES by 17.9% (block-bootstrap CI 6.2–32.4%) and variance by 14.9% (3.9–28.0%), and cut the maximum drawdown from 53.0% to 41.7% with an average notional of 11.1% of the BTC position (Table 3). It was held on 67% of bars and off whenever the estimated spot-vol covariance was positive, including on 8 of the 23 days in the unhedged ES tail. It is a crash hedge (Figure 2). Net of funding, its P&L summed to +5.3% of BTC notional: the ten best days contributed 51% and all other days –46%. The ten worst BTC days cost the unhedged book 74% of notional, and the hedge earned back 36%. The ES reduction was 6.3% in 2024, 10.0% in 2025 and 52.3% in 2026; without the best day it is 11.9%, without the five best 9.0%. Over 7- and 14-day horizons it is 22.2% and 32.4%, so the daily figure is not an artefact of the index lagging across day boundaries. Funding cost the hedge 1.4% of BTC notional a year: it paid 4.6% a year in 2024 and 4.3% in 2025 and received 5.6% in 2026.

Rule	ES red.	Var red.	Max DD	Tail offset	Hedge P&L	Funding	Trading	Turnover	Notional
Unhedged	0.0	0.0	53.0	0.0	0.00	0.00	0.00	0.0	0.0
Always-on (MV)	17.9	14.9	41.7	26.9	2.13	1.41	0.20	1.3	11.1
Always-on, 1.5×	20.1	13.2	41.9	38.7	1.07	2.21	0.29	1.8	16.7
VWAP-switch	18.8	12.6	42.5	25.6	6.70	-0.26	5.17	32.3	4.6
VWAP-ratchet	20.3	15.0	42.6	30.6	1.34	2.67	0.37	2.3	12.5

Table 3: Hedging a 1-BTC spot book with the Bitfinex bviv perpetual, 3 April 2024 to 3 October 2026, executed one bar after the signal at the mark (% , except turnover). ES/Var red.: reduction in daily ES_{97.5} and variance against unhedged. Max DD: maximum drawdown. Tail offset: hedge P&L net of costs as a share of BTC losses on the worst 5% of days. Hedge P&L (index P&L minus funding), Funding and Trading (fees and slippage): % of BTC notional a year. Turnover: perpetual notional traded per year over BTC notional. Notional: average hedge notional in % of BTC.

Always-on hedge: cumulative P&L (% of BTC notional)

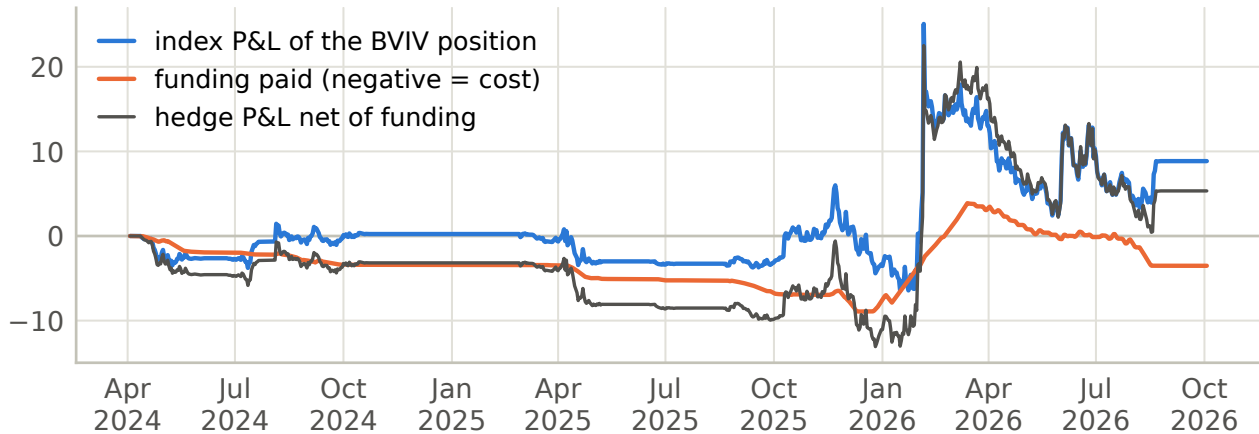


Figure 2: Always-on minimum-variance hedge on live data: cumulative index P&L of the bviv position, funding paid, and hedge P&L net of funding, in % of BTC notional.

The best day. On 5 February 2026 the hedge returned 20% of BTC notional while BTC fell by 14% (Figure 3). bviv stood at 57 at midnight and 61 an hour before BTC broke below the ratchet’s VWAP band on the bar ending 15:15 UTC; it was 67 on that bar, 77 by 15:45 and peaked at 96 at 22:00. All rules entered the day with 558–609 contracts. As the index’s 15-minute moves exploded, the EWMA hedge ratio fell and the rules sold into the spike, the always-on hedge to 205 contracts by the close; the ratchet’s overlay kept it up to 20% larger.

Timing adds mostly size. The switch traded 32.3 times the BTC notional a year; its ES reduction differed from the always-on hedge’s by +0.92 points, not significantly (CI [−2.71, +3.54]), but at the book’s 50 bp per side it would have paid 16.2% of notional a year in trading costs. The ratchet beat the always-on hedge by +2.37 points (CI [−0.22, +4.89]) and the 1.5× hedge by +0.17 (CI [−1.74, +2.35]). Its 129 placebos reduced ES by 19.9% on average, 90% of them between 18.9% and 20.4% (Figure 4, left): +2.04 points of the edge are size and +0.33 timing, and the ratchet beats 76% of its placebos. That rank depends on execution: 91% with same-bar fills, 65% at quoted mids. Across the 72-rule grid, timing adds +6.66 points without a core hedge (rules beat a mean 98% of their placebos), +1.55 with half a core and +0.68 with the full core (84%; 10 of 24 rules beat at least 95%; right panel). VWAP breakdowns time protection far better than chance when nothing else is held, but they flag a sell-off already under way, and a core hedge captures most of their value.

Execution and capacity. Table 4 varies one assumption at a time. Costs, funding and fills at quoted mids barely move the ES reductions, which are dominated by crash days. Speed matters most: same-bar execution would raise the always-on ES reduction to 19.1%, while executing one hour or one day late lowers it to 16.0% and 12.6%. Same-bar fills would also lift the switch’s hedge P&L from +6.70% to +12.80% a year, by collecting the index’s catch-up. Capacity binds. When held, the hedge of one bitcoin was a median 248 contracts, against a median open interest of 87, and exceeded the whole open interest on 61% of those bars (41% of all bars). On its five best days it was 2 to 23 times the open interest; on 5 February 2026 it traded 353 contracts while the market traded 18, and 30% of its trades fell on days without a single trade in the market. A book snapshot on 4 October 2026 shows market makers quoting size: buying or selling one bitcoin’s hedge would have cost 0.50% and 0.50% from the mid (the 0.34% top-of-book spread was dust), and depth within

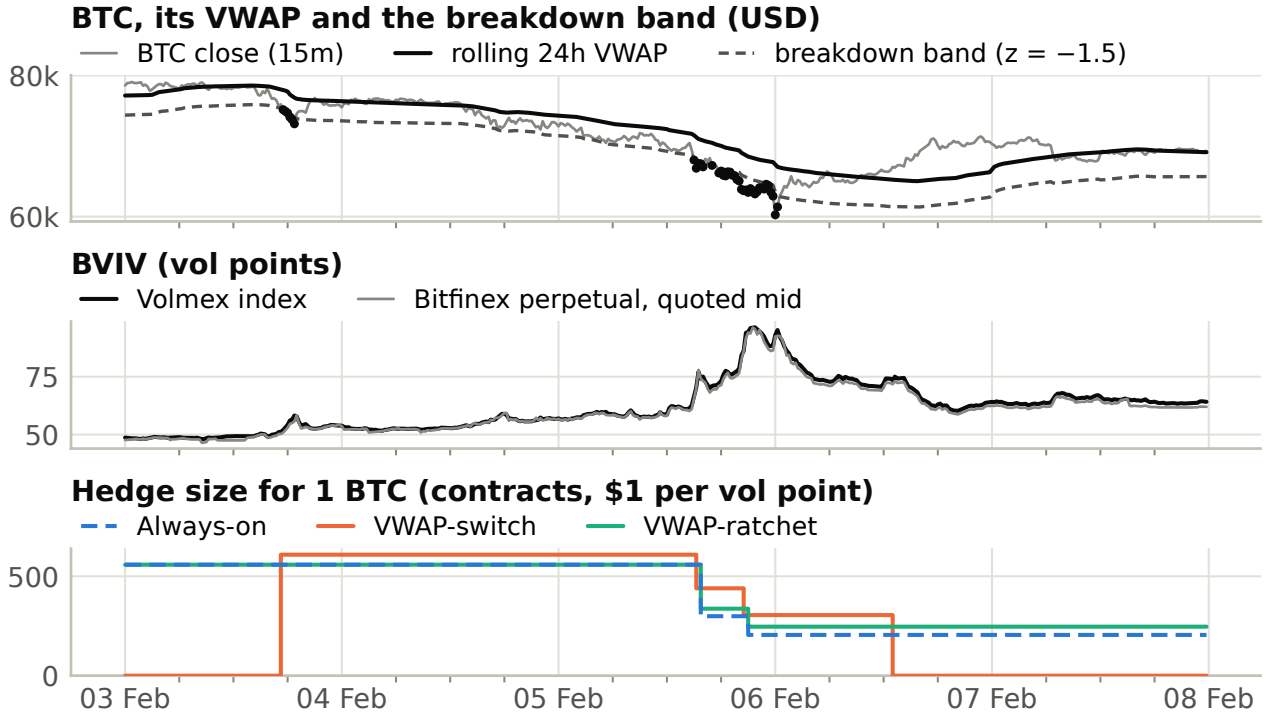


Figure 3: The best hedge day, 5 February 2026, and the days around it at 15-minute resolution (UTC, bar open times). Top: BTC, its rolling 24-hour VWAP and the ratchet’s breakdown band (dots: $z_t < z_{in}$). Middle: the Volmex index and the Bitfinex perpetual’s quoted mid. Bottom: hedge sizes of the three rules for a 1-BTC book.

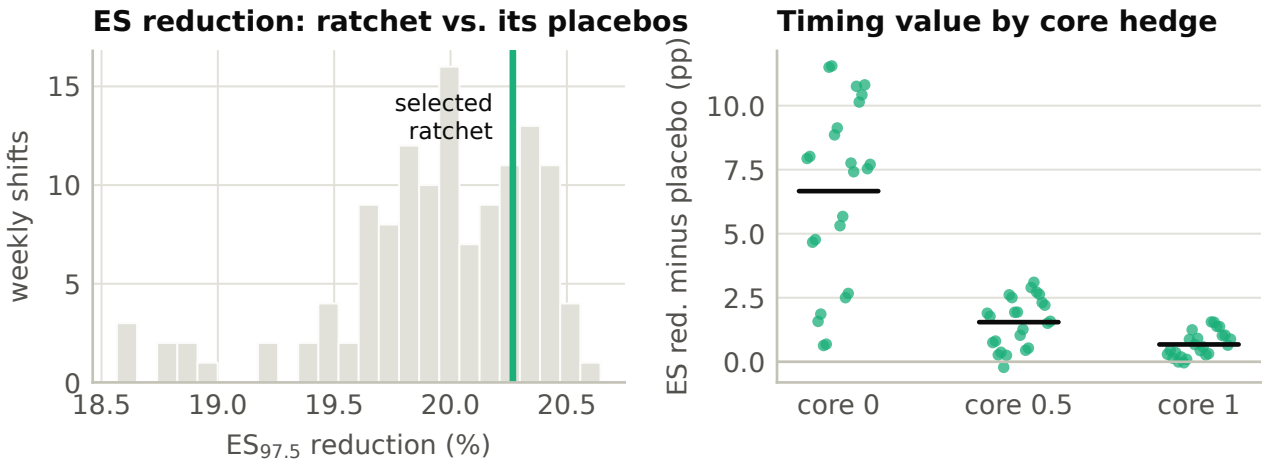


Figure 4: Timing placebos on live data. Left: ES reduction of the selected ratchet (line) against its 129 weekly circular shifts of its triggers. Right: timing value of each of the 72 ratchet configurations (ES reduction minus the mean over 64 two-weekly shifts) by core size; bars mark the means.

2.5% of the mid would have held the hedge of about 4 bitcoins. The measured protection is a benchmark for small books, not a realisable P&L at scale.

6 The controlled experiment

A single path cannot separate rules reliably, so we repeat the study in a simulated market calibrated to the stylised facts (Table 1, last column). Log-variance is the sum of a slow and a fast Ornstein–Uhlenbeck factor; the fast factor tracks a three-state hidden regime (calm, stress, euphoria) and receives EGARCH-type shocks (Nelson, 1991) whose signed loading on the return shock is $-0.30/-0.60/+0.25$ by regime, so volatility rises in sell-offs and, in euphoria, in rallies. Jumps cluster in stress; volatility and volume follow intraday and weekend seasonality. The index is the square root of the model’s exact expectation of 30-day average variance (a Feynman–Kac product over regime paths), scaled by a variance premium and perturbed by persistent sentiment. The perpetual’s funding is the exact one-bar drift transfer plus a carry of 6 vol points a year, so it offers no free lunch from mean reversion; unit tests check both expectations against brute-force simulation. The simulated index has neither the live index’s lag nor a funding cap, so the simulation does

Variant	Always-on	1.5×	Switch	Ratchet
Same-bar execution	19.1 (+2.7)	20.5 (+1.8)	20.1 (+12.8)	21.1 (+3.1)
Execution 1 hour later	16.0 (+0.6)	17.3 (−1.6)	14.1 (+2.2)	18.3 (−1.6)
Execution 1 day later	12.6 (−1.2)	11.2 (−5.1)	10.3 (+0.9)	13.2 (−2.6)
Fills at the quoted mid	17.7 (+2.1)	19.8 (+1.1)	18.4 (+6.7)	20.2 (+1.3)
Costs 50 bp per side	17.8 (+2.1)	20.0 (+1.1)	18.3 (+6.7)	20.1 (+1.3)
No trading costs	18.0 (+2.1)	20.1 (+1.1)	19.0 (+6.7)	20.3 (+1.3)
Funding ignored	17.9 (+3.5)	20.0 (+3.3)	18.7 (+6.4)	20.3 (+4.0)
Perpetual BTC book	18.0 (+2.1)	20.1 (+1.1)	18.9 (+6.7)	20.3 (+1.3)

Table 4: Live-data sensitivities, frozen rules, one change at a time to the baseline (execution one bar after the signal at the mark, 6+10 bp per side): ES reduction in % (hedge P&L in % of BTC notional a year).

Rule	ES red.	Var red.	Max DD	Tail offset	Carry	Trading	Turnover	Time on
Unhedged	0.0	0.0	49.5	0.0	0.00	0.00	0.0	0
Always-on (MV)	4.2	4.1	48.7	5.9	0.74	0.14	0.9	93
Always-on, 1.5×	4.9	3.1	49.5	8.8	1.11	0.20	1.3	93
VWAP-switch	2.8	1.9	50.9	4.9	0.38	3.94	24.6	35
VWAP-ratchet	4.9	3.9	48.9	7.8	0.96	0.23	1.4	98
Ratchet, placebo timing	4.9	3.8	48.7	7.7	0.96	0.24	1.5	98

Table 5: Simulation: hedging a long-BTC book on 200 test paths (medians; % unless stated). ES/Var red., Max DD, Tail offset, Trading and Turnover as in Table 3; Carry: funding carry paid (% of BTC notional a year); Time on: share of days with a hedge. Placebo timing: the selected ratchet with circularly shifted triggers (mean of 5 shifts).

not speak to execution delay or the funding regime. Each path has 900 days; rules are selected on 48 training paths and evaluated on 200 test paths.

Findings. The forecast ranking is the same as on live data: HAR-IV is best at one day (median QLIKE ratio 0.95) and seven days (0.89), the bias-corrected index at 30 days (0.85). The hedging results have the live pattern at a smaller scale (Table 5), because the simulated spot-vol correlation (−0.21) is weaker than in 2025–26. The always-on hedge reduces ES by 4.2%; the switch trails it by 1.67 points (CI [−2.00, −1.36]). The ratchet beats it by +0.53 points (CI [+0.33, +0.72]) but its placebo by only +0.05, and a 1.5× hedge reaches the same ES reduction (4.9%): as on live data, the edge is mostly size. Across the grid, timing adds +0.78 points without a core and +0.02 with the full core. In 96 further paths with common random numbers, the ratchet’s edge survives doubled or halved trading costs and a carry of 0 to 24 vol points a year (+0.63 to +0.67 points). It shrinks to +0.18 points under inverse leverage and to +0.34 when crashes cluster in stress, where the CIs include zero.

7 Limitations and conclusion

The live sample is one path of 914 days whose tail is dominated by a few crashes in 2026, which the bootstrap quantifies but cannot remove. Fills one bar after the signal at the mark plus costs are a benchmark: on its best days the market’s volume and open interest could not have absorbed even a 1-BTC hedge, and we observe the order book only once, after the sample. The gains also assume that the shorts stayed solvent through a 69% intraday rise of the index and that USDt held its peg; we do not model margin, liquidation, auto-deleveraging or venue failure. Oracle risk on builder-deployed venues (Hyperliquid, 2026) and algorithmic flow at the 15-minute boundary (Kim and Hansen, 2026) argue for working orders inside the bar, e.g. by VWAP slicing (Berkowitz et al., 1988; Konishi, 2002; Białkowski et al., 2008). The simulation’s spot-vol dynamics are calibrated, not estimated.

Four lessons follow. (1) HAR augmented with implied variance is the best one-day BTC volatility forecast; at a week and a month HAR-IV and the bias-corrected implied index lead HAR, but not significantly. (2) A minimum-variance BVIV hedge cut tail risk by 17.9% over 2024–26, but by 6.3% in 2024, 10.0% in 2025 and 52.3% in 2026: the protection rests on a few crashes with spiking implied volatility. Holding beats switching on costs and in the simulation; on live data the two cannot be told apart. (3) Its price is funding, which on Bitfinex is capped, volatile and usually paid by longs; its limits are execution speed and capacity. (4) A 15-minute VWAP breakdown is real but secondary information: without a core hedge it beats random timing, but with one most of a ratchet’s edge is size; timing adds +0.33 points for the frozen rule and +0.68 across the full-core grid. Intraday timing rules should be split into size and timing against such placebos before they earn their complexity.

References

- Damien Ackerer, Julien Hugonnier, and Urban Jermann. Perpetual futures pricing. *Mathematical Finance*, 2025. doi: 10.1111/mafi.70018.
- Carol Alexander and Arben Imeraj. The Bitcoin VIX and its variance risk premium. *Journal of Alternative Investments*, 23(4):84–109, 2021. doi: 10.3905/jai.2020.1.112.
- Carol Alexander, Dimitris Korovilas, and Julia Kapraun. Diversification with volatility products. *Journal of International Money and Finance*, 65: 213–235, 2016. doi: 10.1016/j.jimonfin.2015.10.005.
- Caio Almeida, Maria Grith, Ratmir Miftachov, and Zijin Wang. Risk premia in the Bitcoin market. arXiv preprint 2410.15195, revised August 2025, 2024.
- Torben G. Andersen and Tim Bollerslev. Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2–3): 115–158, 1997. doi: 10.1016/S0927-5398(97)00004-2.
- Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003. doi: 10.1111/1468-0262.00418.
- David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5):458–471, 2014. doi: 10.1090/noti1105.
- Dirk G. Baur and Thomas Dimpfl. Asymmetric volatility in cryptocurrencies. *Economics Letters*, 173:148–151, 2018. doi: 10.1016/j.econlet.2018.10.008.
- Lykke Ø. Bergsli, Andrea F. Lind, Peter Molnár, and Michał Polasik. Forecasting volatility of Bitcoin. *Research in International Business and Finance*, 59:101540, 2022. doi: 10.1016/j.ribaf.2021.101540.
- Stephen A. Berkowitz, Dennis E. Logue, and Eugene A. Noser, Jr. The total cost of transactions on the NYSE. *Journal of Finance*, 43(1):97–112, 1988. doi: 10.1111/j.1540-6261.1988.tb02591.x.
- Jędrzej Białkowski, Serge Darolles, and Gaëlle Le Fol. Improving VWAP strategies: A dynamic volume approach. *Journal of Banking & Finance*, 32(9):1709–1722, 2008. doi: 10.1016/j.jbankfin.2007.09.023.
- Elie Bouri, Georges Azzi, and Anne Haubo Dyhrberg. On the return-volatility relationship in the Bitcoin market around the price crash of 2013. *Economics: The Open-Access, Open-Assessment E-Journal*, 11(2017-2), 2017. doi: 10.5018/economics-ejournal.ja.2017-2.
- Peter Carr and Liuren Wu. Variance risk premiums. *Review of Financial Studies*, 22(3):1311–1341, 2009. doi: 10.1093/rfs/hhn038.
- CME Group. CME Group to launch Bitcoin volatility futures contracts. Press release, May, 2026.
- CoinDesk. Bitcoin volatility index’s correlation with price flips positive, boosts appeal of bullish call options. CoinDesk, 4 April, 2023.
- CoinDesk. Bitfinex derivatives users can now place bets on bitcoin and ether implied volatility. CoinDesk, 3 April, 2024.
- CoinDesk. Volmex’s bitcoin and ether volatility futures top \$10M volume since debut as traders look beyond price. CoinDesk, 25 July, 2025a.
- CoinDesk. Bitcoin’s record rally from \$70K to \$118K: A tale of transition from wild west to wall street-like dynamics. CoinDesk, 11 July, 2025b.
- CoinDesk. Perp futures linked to ‘bitcoin VIX’ debut on Hyperliquid. CoinDesk, 21 September, 2026.
- Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009. doi: 10.1093/jfinec/nbp001.
- Kresimir Demeterfi, Emanuel Derman, Michael Kamal, and Joseph Zou. A guide to volatility and variance swaps. *Journal of Derivatives*, 6(4):9–32, 1999. doi: 10.3905/jod.1999.319129.
- Deribit. DVOL – Deribit implied volatility index. Deribit Insights, exchange documentation, 2021.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.
- Louis H. Ederington. The hedging performance of the new futures markets. *Journal of Finance*, 34(1):157–170, 1979. doi: 10.1111/j.1540-6261.1979.tb02077.x.
- Peter Reinhard Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005. doi: 10.1198/073500105000000063.
- Peter Reinhard Hansen, Chan Kim, and Wade Kimbrough. Periodicity in cryptocurrency volatility and liquidity. *Journal of Financial Econometrics*, 22:224–251, 2024. doi: 10.1093/jfinec/nbac034.
- Songrun He, Asaf Manela, Omri Ross, and Victor von Wachter. Fundamentals of perpetual futures. arXiv preprint 2212.06888, 2022.
- Lai T. Hoang and Dirk G. Baur. Forecasting Bitcoin volatility: Evidence from the options market. *Journal of Futures Markets*, 40(10):1584–1602, 2020. doi: 10.1002/fut.22144.
- Hyperliquid. Funding; HIP-3 builder-deployed perpetuals. Hyperliquid documentation, 2026.
- Roni Israelov. Pathetic protection: The elusive benefits of protective puts. *Journal of Alternative Investments*, 21(3):6–33, 2019. doi: 10.3905/jai.2018.1.066.
- Leland L. Johnson. The theory of hedging and speculation in commodity futures. *Review of Economic Studies*, 27(3):139–151, 1960. doi: 10.2307/2296076.
- J.P. Morgan/Reuters. *RiskMetrics – Technical Document*. J.P. Morgan/Reuters, New York, 4th edition, 1996.
- Chan Kim and Peter Reinhard Hansen. The quarter-hour effect: Periodic algorithmic trading and return predictability in cryptocurrency futures. arXiv preprint 2607.09426, 2026.
- Hizuru Konishi. Optimal slice of a VWAP trade. *Journal of Financial Markets*, 5(2):197–221, 2002. doi: 10.1016/S1386-4181(01)00023-4.
- Lily Y. Liu, Andrew J. Patton, and Kevin Sheppard. Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1):293–311, 2015. doi: 10.1016/j.jeconom.2015.02.008.
- Jacob A. Mincer and Victor Zarnowitz. The evaluation of economic forecasts. In Jacob A. Mincer, editor, *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, pages 3–46. NBER, 1969.
- Daniel B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991. doi: 10.2307/2938260.
- Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.
- Edward Szado. VIX futures and options: A case study of portfolio diversification during the 2008 financial crisis. *Journal of Alternative Investments*, 12(2):68–85, 2009. doi: 10.3905/JAI.2009.12.2.068.
- Tetsuya Takaishi. Time-varying properties of asymmetric volatility and multifractality in Bitcoin. *PLoS ONE*, 16(2):e0246209, 2021. doi: 10.1371/journal.pone.0246209.
- Volmex Labs. Volmex implied volatility indices: Methodology. White paper, <https://volmex.finance/Volmex-IV-paper.pdf>, n.d. Accessed October 2026.
- Robert E. Whaley. Trading volatility: At what cost? *Journal of Portfolio Management*, 40(1):95–108, 2013. doi: 10.3905/jpm.2013.40.1.095.
- A. Elizabeth Whalley and Paul Wilmott. An asymptotic analysis of an optimal hedging model for option pricing with transaction costs. *Mathematical Finance*, 7(3):307–324, 1997. doi: 10.1111/1467-9965.00034.
- Halbert White. A reality check for data snooping. *Econometrica*, 68(5): 1097–1126, 2000. doi: 10.1111/1468-0262.00152.